Introduction to Computer Science

Diploma in software Engineering 24.3

Coursework – Software Engineering

CODSE24.3F-110 – S.H.A.I. FERNANDO



SCHOOL OF COMPUTING AND ENGINEERING

NATIONAL INSTITUTE OF BUSINESS MANAGEMENT

COLOMBO - 07

❖ Specification of a Desktop Computer

- **Casing - Tower**

  Computer casing is a protective case for the internal components of the computer there are different types of cases such as: - Full tower

  Mid tower

  Mini case

  Micro case

  And also when choosing a tower casing we need to check it`s ventilation and there should be enough space inside the casing for good air flow so that it reduce the overheating.

- **Processor – intel core i7 11th gen**

  Processor is the brain of the computer. It performs many program tacks and the processor gets instructions from the RAM. Main functions of the processor are,

  - Fetch and execute commands.
  - Temporally store data being processed.
  - Carry out all arithmetic calculation and does the logic operations.

  Intel care i7 11th gen has total cores of 8 which help full to do 8 individual tasks at the same time, and total 16 threads which the processor can handle 16 simultaneous tasks, and also, 16MB cache memory, which gives a big cache size to make the data access higher.

- **Chipset – Q570**

  Motherboard is a circuit board inside computer that stores electrical components and chipset are the electrical components of the motherboard. Intel Q570 chipset support 10th gen and 11th gen and also support dual-channel DDR4 memory up to 64GB. So, intel Q570 chipset has a good power to do powerful tasks.

- **Operating system – windows 11 pro**

  A type of system software that manage computer hardware and software. And windows 11pro is a very good operating system launched by Microsoft`s and there are more advanced features in 11 pro which are not found from home edition of windows 11 such as good control, security and management in tools.

- **RAM – 16GB**

  RAM is a type of computer memory that stores data temporarily when the computer is running. 16GB DDR4 3200MHz is commonly used in modern desktops, 16GB of capacity of a memory is good for gaming and other hard works done by the pc. DDR4 is it's a 4th gen of DDR memory. So, it is fast and more powerful. 3200MHz is the speed of its reading data.

- **DIMM slots – 4**

  DIMM slots are found on your mother board they are used to hold ram sticks. Also, there are different types of DIMM slots for DDR 3, DDR 4 etc. In good powerful mother

board, it comes with 4 DIMM slots. There are mother boards comes with more than 4 DIMM slots also.

- **SSD – 512GB**

  An SSD is a secondary storage device that store data permanently. SSD are faster than HDD so can get data fast there are different storage capacities 128GB, 256GB, 512GB, 1TB etc. commonly 512 are used in most of the desktops because it can store lot of stuff like games, videos, documents etc.…

- **HDD – 1TB 7200rpm**

  An HDD is a secondary storage device that store data permanently. HDD are slow compare to SSD but can used to store data with more capacity, 1TB with 7200rpm SATA HDD is a good storage device that can used for storing large files with a good transfer speed.

- **WLAN – 802. 11ac**

  WLAN is a method of wireless distributing network connectivity to multiple device within a local area. It enables communication among computers and other devices using radio transmissions. WLAN 802.11ac is a good network with more improved features and with new technologies.

- **USB version 3**

  USB port is used to connect with smartphones, computers and other electronic devices. USB version 3 can read or write data faster. Now a days most of the desktops come with USB version 3. Also, modern computers come with more than 3 USB ports.

- **USB type C**

  This is the port that can see in the most of the modern electronical devices. USB type C data transfer speed is high and USB type C cables are smaller and eliminate the need for mini USB ports.

- **HDMI**

  HDMI stands for High Definition Multimedia Interface. It is most frequently used to transfer audio and visual content from one device to another device. From this can get a better-quality output. Most of the modern desktops use this instead of VGA port.

- **EXTERNAL PORTS – RJ45**

  External ports are the ports that we can see in the rear of the casing. To those are the external devices connected using cables. These external ports are used to connect peripherals such as RJ45 cables etc.… RJ45 port is used to connect WIFI or internet connection to the desktop.

- **POINTING DEVICE**

  A pointing device is a hardware input device that allows the user to move the mouse courser in the computer program to point or click. For example, some pointing devices are mouse, joystick, light pen, touch pad.

- **KEYBOARD – FULL SIZE**

  A keyboard is a primary input device use in computer to input Letters, Numbers, symbols etc.… There are different types of keyboard from those keyboard full size keyboards are the best because it has 104 keys and also have main cluster keys, the main function keys, arrow keys and home cluster keys.

- **DISPLAY – 23.8 LED**

  This display also known as monitor. This is an output device that display the graphics information generated by the computer GPU also used to display images, text, videos etc.… 23.8 inch also known as 24 inch and the display of the monitor built by LED panel so we can see realistic colors from this LED monitor.

- **WARRENTY – 3 YEARS**

  A warranty is a guarantee given by a manufacture party or a seller according to their product. Most of the desktop parts and laptops get 3 years warranty for their product. And also, they give a labor warranty which also means service warranty. On their warranty time period they give their service for their product free of charge.

❖ A small digital marketing agency

This agency does website management, content creation, graphic design and social media marketing.

- **Hardware specification for desktops** - use for content creation, graphic designing.

    - **Processor: AMD Ryzen 5 5600g**

      This AMD Ryzen 5 5600g processor is a good balance processor. This processor is powerful to run all design tools smoothly and also not too expensive.

    - **Memory: 16GB**

      16GB ram can do tasks faster and smoothly and can run multiple task at the same time with out getting crash.

    - **Storage: SSD 512GB**

      SSD are better than HDD because SSD are faster and smaller than HDD. 512GB of storage is very useful to store more details and also can store large files without any issue.

    - **Graphics card: RTX 3060ti 6GB**

      From this VGA card you can do any heavy work smoothly. Content creations, graphic designs can be done without any issue. Also, it has 6GB of V RAM there for can do heavy multiple tasks.

- **Hardware specification for laptops** – use by employee who need flexibility and employees who work from home.

    - **Processor: AMD Ryzen 5 5600g**

      Ryzen 5 5600g processor gives a good performance that can handle heavy work too from this processor it can do content creations, website creations easily. Also, a budget friendly processor.

    - **Memory: 16GB**

      16GB RAM is good when working with heavy webpages and video creations. Because of 16GB can run apps very smoothly.

    - **Storage: SSD 512GB**

      SSD can access information fast so that works get easier and can store large files because of 512GB space. This kind of large storage is needed when working with large files.

    - **Graphic card: RTX 2080 8GB**

      With using this RTX 2080 can run heavy works and can do any video editing or graphic designing part without getting crash and also has a V RAM 8GB so can run heavy applications smoothly.

- **Hardware specification for servers –** use for maintain websites and to manage client data.

    - **Processor: Intel XEON gold**

      This processor does multi – core processing. This processor is making to do heavy works for example website maintain.

    - **Memory: 32GB**

      This server needs more RAM to handle multiple tasks. 32GB RAM can have smooth file accessibility. But because of heavy load work may be need to go for 64GB so that the heavy load work can be done very easily.

    - **Storage: 2TB SSD**

      Server store lot of details of website and also store details about clients so need a bigger storage.

    - **Network card: Gigabyte Ethernet**

      Fast network is need to handle large files and to transfer quickly and need to build smooth connection. Gigabit Ethernet is good for small businesses but if data traffic happens need to move for 10GB ethernet.

❖ Operating systems and application software used by a small digital marketing agency.

- **Operating systems**

  - **Windows:** windows is the operating system used by many it has capability of running different software tool that are needed for a digital marketing agency. For example, it can run Microsoft office, Adobe etc.…

  - **macOS:** This operating system has a user- friendly interface. This is very popular among designers and content creators, because of its easiness to work with design software like photoshop, lightroom, Adobe etc.…

- **Application software**

  - **Microsoft office 365**: Tools like word, PowerPoint, excel these tools are used to create spreadsheets, manage emails and document creation.
  - **Buffer:** buffer is used to manage multiple accounts, manage posts and also to analyses performance.

- **Adobe:** This application used for design, video editing and content creations.
- **Google analytics:** This application software used for tracking website traffic.
- **Google drive:** can store flies in cloud and also gives workspace in google apps.

❖ Network devices used by a small digital marketing agency

- **Switch:** switches are the main building units of any network. Switch helps to connect multiple devices such as computers, printers and servers within the same building.
- **AP:** The AP is known as Access point. An access point is connected directly to a wired LAN, that provide connection point for wireless users.
- **Router:** Router connects the agency`s local network to the internet and provide security features such as firewall.
- **Modem:** Modem translates digital signals from the agency router into a format that can be transmitted over cable lines.
- **Firewall**: Protects the agency`s network from external threats like malwares.

❖ 10 Best practices to have a better lifetime for your pc

1. Keeping the PC clean - Daily cleaning your pc both inside and outside to prevent your pc from dust.
2. Software update - by keeping your pc with the latest software update to get best performance and security from the operating system.
3. Not letting to overheat - using cooling fans to maintain the temperature inside your pc and also choosing a good casing which has a good air flow.
4. Cleaning disk space - cleaning up unwanted files, and keep the disk clean to prevent slowdowns.
5. Avoid physical damages - keeping the pc in a safe location and also away from liquids.
6. Update device drivers - by updating all hardware drivers to keep them in good health and to take best performance.
7. Protect from malware - by using a virus card and keeping it update can protect your pc from some malwares.
8. Using external storage - storing large files in external storages so pc won't get slow.

9. Shut downing correctly - shut downing correctly can keep your pc hardware part longer with a good health.
10. Avoiding overclocking - overclocking makes components to pass their limits so that leads to overheating.

❖ 10 Best practices to keep your pc safe and secure

1. Using strong password - creating strong passwords by using letters, numbers and symbols.
2. Install antivirus software - using antivirus software to protect your pc from different virus.
3. Backup important data - If something happen to the system or a cyberattack no need to worry you can get you data through backup.
4. Use encryption - If your system is compromised, the data remains unreadable for unknown users.
5. Activate automatic locking - setting your computer to lock after a certain time when your away from it.
6. Delete unnecessary files - deleting unused, unwanted files to reduce the risk of malware.

7. Install software from trusted websites – download software only from official websites to get ride of unwanted malware.
8. Be aware of USB device - Not using untrusted USB drive because they can spread malware.
9. Enable firewall - firewall is active to block unauthorized access to your computer.
10. Avoid phishing scans - Avoid entering personal information on suspicious websites.

❖ Understanding Incremental and Differential backups

Backups are very essential for protecting data, in case if system failures, data corruption or accidental deletion happen the data can be restored without any difference. There are different types of backups, Incremental and Differential backups are commonly used methods for backup restore. And also, there are some advantages and disadvantages in both backups.

- **Incremental backup**

  An incremental backup saves only the data that has changed since the last backup. This means that after the initial full backup, each incremental backup will only capture the data that has been modified since the previous backup.

- Advantages of Incremental backup

  - ✓ Faster backup: As only the modified data is backups, the time taken for backups is very low.
  - ✓ Lower bandwidth: It was less network bandwidth for backups due to small file size.
  - ✓ Efficient storage: It requires less storage space than differential backups.

- Disadvantages of Incremental backup

  - ✓ Higher restore time: you need to sequentially restore the backup and each incremental backup, restoration is time consuming.
  - ✓ Backup dependency: if one backup in the chain is lost the hole incremental backups become useless.
  - ✓ Restore complexity: to restore data, the full backup all incremental backups since that full backup are required. This cause to slower recovery time and a higher risk of errors.

- **Differential backup**

A differential backup saves all data the has changes since the last full backup. Not like incremental backups. It only backup changes since the last backup differential backups always refer back to the last full backup.

- Advantages of Differential backup

  - ✓ Fast restoration: only the full backup and the last differential backup are needed. So, this is fast and not complex.
  - ✓ Easy restore process: There is only one differential backup. There is less chance of losing data during restoration.
  - ✓ Less backup dependency: only depend on the full backup and the latest differential backup.

- Disadvantages of Differential backup

  - ✓ Longer backup time: the differential backup grows with each day. So, the time taken to create a differential backup increase.
  - ✓ High bandwidth: differential backups require more bandwidth due to large backup size.
  - ✓ Storage: differential backups can become large because the record all changes since the last full back up so it consumes more storage.

❖ Summary

This report outlines a critical hardware-software configuration, optimized for a set of professional applications with respect to desktop computers, laptops, and servers pertaining to the fields of digital marketing and graphic design. Advanced specification of multi-core processors, high-capacity SSDs, and dedicated GPU has been highlighted to become focused on performance, multitasking, and data storage efficiency. Further, network infrastructure components are discussed, including Gigabit Ethernet and WLAN standards, to ensure connectivity without any errors. After that, it describes the concept of both incremental and differential backups, considering speed-bandwidth-restoration complexity tradeoffs. Proactive maintenance activities summarized here in include thermal management, driver updating, and cybersecurity for the purpose of keeping the system up and operational.